

CliniScan: Chest X-ray Lung Abnormality Detection System

Milestone 2: Object Detection Model Development & Baseline Training
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1. Project Overview & Objective

The objective of this milestone is to develop a state-of-the-art deep learning object detection model capable of localizing and identifying abnormalities in Chest X-ray images.

Unlike a simple classification system, this object detection system predicts bounding boxes to precisely locate: - **Abnormalities** (as a unified localized class)

This model acts as the core stage in the automated medical diagnosis pipeline, helping radiologists precisely pinpoint anomalous regions for further analysis.

2. Dataset Preparation

The dataset used for training contains chest X-ray images and expert bounding box annotations from the VinBigData Chest X-ray dataset. Images and annotations were preprocessed and organized to train a YOLO (You Only Look Once) detection model.

2.1 Image Preprocessing

The following preprocessing steps were applied: - Conversion of raw DICOM medical images to standardized PNG format. - Image resizing to 512×512 pixels to preserve detailed anatomical structures. - Conversion of absolute bounding box coordinates to YOLO format (normalized `x_center`, `y_center`,

`width`, `height` scaling from 0.0 to 1.0). - Data augmentation (mosaics, scaling, shifting) applied automatically via YOLO dataloaders for improving model generalization.

These steps ensure consistent input morphology and prevent format-related training errors.

2.2 Dataset Organization

The dataset was organized into parallel `images` and `labels` directories conforming to the Ultralytics YOLO dataset specification.

Folder Structure

```
classification_data/

  images/
    train/
    val/

  labels/
    train/
    val/
```

This structure, mapped via a `dataset.yaml` file, allows the PyTorch-based YOLO data loader to pair every image with its corresponding text file of bounding boxes.

3. Object Detection Model (YOLOv8x)

The detection model is based on **YOLOv8x (Extra Large)**, an advanced, highly accurate, single-stage object detection network developed by Ultralytics.

Transfer learning was applied using pre-trained weights to adapt the model to recognize chest X-ray abnormalities instead of standard everyday objects.

Model Architecture		Component		Specification			:—		:—			Base Model								
		YOLOv8x (Extra Large)			Pretrained Weights		COCO Dataset			Input Size		512 × 512 RGB			Output Classes		1 (Abnormality)			Framework
		PyTorch / Ultralytics																		

The pretrained extra-large backbone provides massive parameter depth, allowing the model to decipher complex and subtle lung opacities and lesions.

4. Training Configuration

The following parameters were utilized in the hyper-parameter configuration during local fine-tuning.

Parameter	Value
Loss Function	Box Loss + Class Loss + DFL (Distribution Focal Loss)
Optimizer	AdamW / Auto
Image Size (imgsz)	512 × 512
Batch Size	4 (Lowered to prevent CUDA OOM)
Epochs	25
Device	GPU (device=0)

The model was iteratively trained using a dynamic learning rate to optimize Mean Average Precision (mAP) for bounding box predictions.

5. Results & Performance

After training, the YOLOv8x detection model achieved significant improvements on the validation dataset.

Performance Metrics - Validation mAP50: Achieved stable precision-recall metrics demonstrating consistent object bounding box accuracy. - **Training Box Loss:** Decreased continuously across all 25 epochs indicating learning of spatial offsets. - **Validation Class Loss:** Settled as the model effectively realized it was detecting the single target ('Abnormality').

The results indicate that the model can successfully overlay accurate bounding boxes around visual signatures of pulmonary anomalies.

6. Challenges & Observations

During model development, several technical challenges were encountered and resolved: 1. **Label Coordinate Corruption:** Bounding box anomalies mathematically exceeding 1.0 or dropping below 0.0 necessitated the creation of data-cleansing script fixes. 2. **Hardware Limitations (VRAM):** Training the massive YOLOv8x model on full-scale images caused CUDA Out-of-Memory (OOM) errors. The batch size was consequently reduced to 4 to ensure stable epoch completion. 3. **Variable Opacities:** Chest X-rays contained complex overlapping bone structures that occasionally obscured mild localized abnormalities, requiring strong spatial attention from the neural network.

Despite these hurdles, data cleansing and VRAM optimization allowed the model to train successfully.

7. Conclusion

In Milestone 2, an advanced YOLOv8x-based object detection model was successfully trained to actively localize abnormalities in chest X-ray images.

By leveraging PyTorch, the Ultralytics framework, and a carefully processed VinBigData dataset, the model correctly predicts bounding coordinates with notable validation accuracy.

This localization capability lays a significantly stronger foundation than holistic classification, paving the way for multi-class medical diagnostic workflows in future project milestones.